ModelIt!

Student Activity Pack

Why Some People Get Addicted and Others Don't

Modeling Neurotransmitter Systems, Genetic Vulnerability, and Addiction Pathways

Grade 12L1 | NGSS HS-LS1-2, HS-LS3-1 | Cause and Effect

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Why do you think some people become addicted to substances while others who try the same thing don't?**

2. **Do you think addiction is a choice, a disease, or something else? Explain your reasoning.**

3. **What do you know about how drugs affect the brain? Be as specific as you can.**

4. **What do you think is more effective for preventing addiction: scaring people or educating them? Why?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Genetic Vulnerability | Substance Exposure | Dopamine Surge | Receptor Downregulation | Addiction Severity

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Low Risk

Genetic Vulnerability: Low | First Exposure: Age 25 | Stress: Low

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: High Risk

Genetic Vulnerability: High | First Exposure: Age 14 | Stress: Moderate

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Stress Amplification

Genetic Vulnerability: Moderate | First Exposure: Age 17 | Stress: Chronic high

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Prefrontal Cortex Development — The brain region responsible for impulse control and long-term planning — not fully mature until age 25, explaining why adolescent substance use carries higher addiction risk than adult use

Social Environment — Peer influence, family support, community connectedness, and access to substances — can either amplify or buffer biological vulnerability depending on quality

Epigenetic Modification — Changes in gene expression caused by environmental factors without altering DNA sequence — stress and substance exposure can activate or silence addiction-related genes, potentially across generations

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Two people try the same substance — one walks away and never thinks about it again, while the other develops a devastating addiction. If the drug is the same, what's different about the people? Use evidence from your simulation to support your answers.

1. **How does understanding the neuroscience of addiction change the way you think about people struggling with substance use disorders?**

2. **Why does the age of first substance exposure matter so much according to your model?**

3. **If genetic vulnerability accounts for 40-60% of addiction risk, what does that mean for prevention strategies?**

4. **How should the criminal justice system treat addiction differently if it's a brain disease rather than a moral choice?**

5. **What are the ethical implications of genetic testing for addiction vulnerability?**

8 STEM Challenge: Design an Evidence-Based Prevention Program

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YOUR ENGINEERING MISSION

Design a science-based addiction prevention program for high school students that uses neuroscience evidence rather than fear tactics, addressing both biological risk factors and environmental protective factors.

THE SCENARIO

Your school district is replacing its outdated 'Just Say No' program with an evidence-based approach. Research shows that fear-based programs are ineffective and can actually increase curiosity. The district wants a program that teaches students the neuroscience of addiction, helps them assess their own risk factors, and builds evidence-based protective skills.

Engineering Design Thinking

1. WHAT NEUROSCIENCE CONCEPTS DOES YOUR PROGRAM TEACH AND HOW ARE THEY PRESENTED TO BE ENGAGING WITHOUT BEING PREACHY?

2. HOW DOES YOUR PROGRAM ADDRESS GENETIC VULNERABILITY WITHOUT CAUSING ANXIETY IN STUDENTS WHO MAY HAVE FAMILY HISTORY?

3. WHAT SPECIFIC PROTECTIVE SKILLS DOES YOUR PROGRAM BUILD AND WHAT EVIDENCE SUPPORTS THEIR EFFECTIVENESS?

4. HOW WOULD YOU EVALUATE WHETHER YOUR PROGRAM ACTUALLY REDUCES SUBSTANCE USE INITIATION?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Neuroscientists studying addiction research how drugs change brain structure and function, earning $70,000-$130,000/year. Addiction counselors help people recover using evidence-based therapies, earning $40,000-$65,000/year. Pharmacologists develop medications to treat addiction by targeting specific receptor systems, earning $80,000-$150,000/year.